



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3213 – BUSINESS INTELLIGENCE

SECTION 01

PHASE 3: VISUALIZATION AND DASHBOARDS

Strategic Real Estate Investment Intelligence: A Multi-Dimensional Analytics Solution for Property Valuation and Market Opportunity Mapping

LECTURER: TS. DR. NOR AZIZAH ALI

GROUP PACMAN

STUDENT NAME	MATRIC NO
LIM YU HAN	A23CS0241
NAJMA SHAKIRAH BINTI SHAHRULZAMAN	A23CS0140
NAWWARAH AUNI BINTI NAZRUDIN	A23CS0143

1.0 Dashboard 1: Market & Affordability Ratio

Market & Affordability Analysis

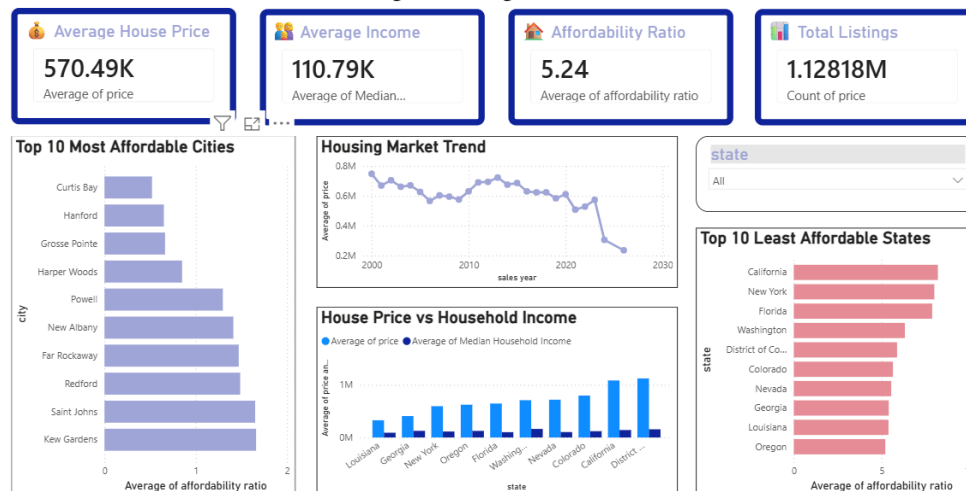


Figure 1: Market and Affordability Analysis Dashboard

Objective

The objective of this dashboard is to analyse housing affordability across different locations by comparing property prices with household income levels. It supports the project's goal of identifying potential real estate investment opportunities through affordability analysis and market trend evaluation.

Dashboard Description

Figure 1 shows the Market & Affordability Analysis dashboard. The dashboard provides an overview of the housing market through four KPIs which are Average House Price (570.49K), Average Household Income (110.79K), Affordability Ratio (5.24), and Total Listings (1.13 million).

The dashboard identifies the Top 10 Most Affordable Cities and Top 10 Least Affordable States using the Affordability Ratio feature. A lower affordability ratio indicates that housing prices are more aligned with local income levels, while a higher ratio suggests lower affordability.

Furthermore, the Housing Market Trend visual shows changes in average house prices over time, while the House Price vs Household Income visual compares property values and income levels across different states. There is also a state slicer to allow users to interactively explore affordability patterns and market conditions in specific regions.

2.0 Dashboard 2: Property Characteristic Analysis

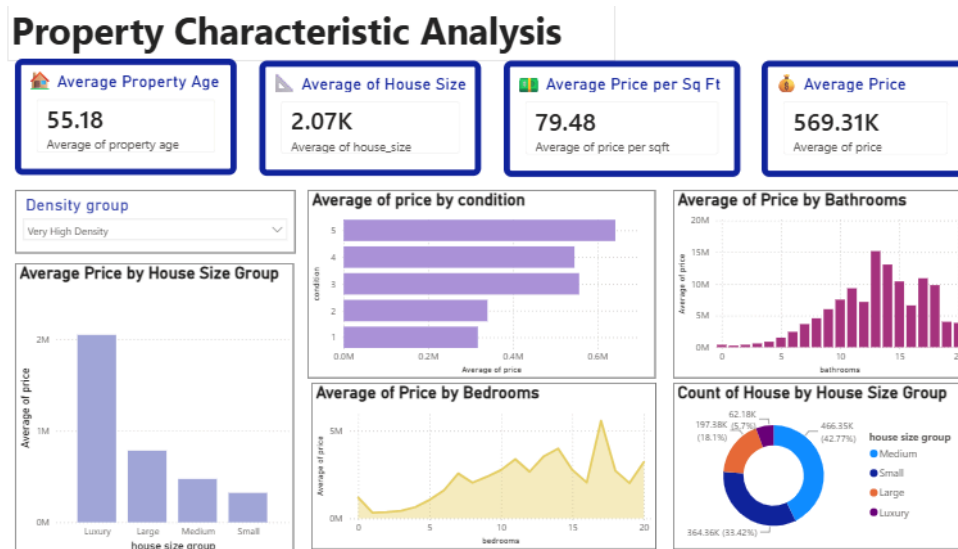


Figure 2: Property Characteristic Analysis Dashboard

Objectives

The objective of this dashboard is to analyze the characteristics of residential properties and identify the factors that influence housing prices. It provides insights into property age, house size, price per square foot, selling price and how different property attributes such as bedrooms, bathrooms, house size groups, density groups, and property condition affect the average house price. These insights help stakeholders understand market trends and support better real estate decision-making.

Dashboard Description

Figure 2 presents the Property Characteristic Analysis dashboard, which provides an overview of key property features and their relationship with housing prices. The dashboard displays four key performance indicators (KPIs): Average Property Age (55.18 years), Average House Size (2.07K sq ft), Average Price per Square Foot (79.48), and Average House Price (569.31K).

The visualizations examine how different property characteristics influence house prices. These include the average price by house size group, property condition, number of bathrooms, and number of bedrooms, allowing users to identify trends and compare property values based on different features.

Additionally, the dashboard includes a Count of Houses by House Size Group donut chart to show the distribution of properties across size categories. A Density Group slicer enables users to filter the data by residential density, making it easier to compare pricing patterns across different neighbourhood types.

3.0 Dashboard 3: Geographic and Density Distribution

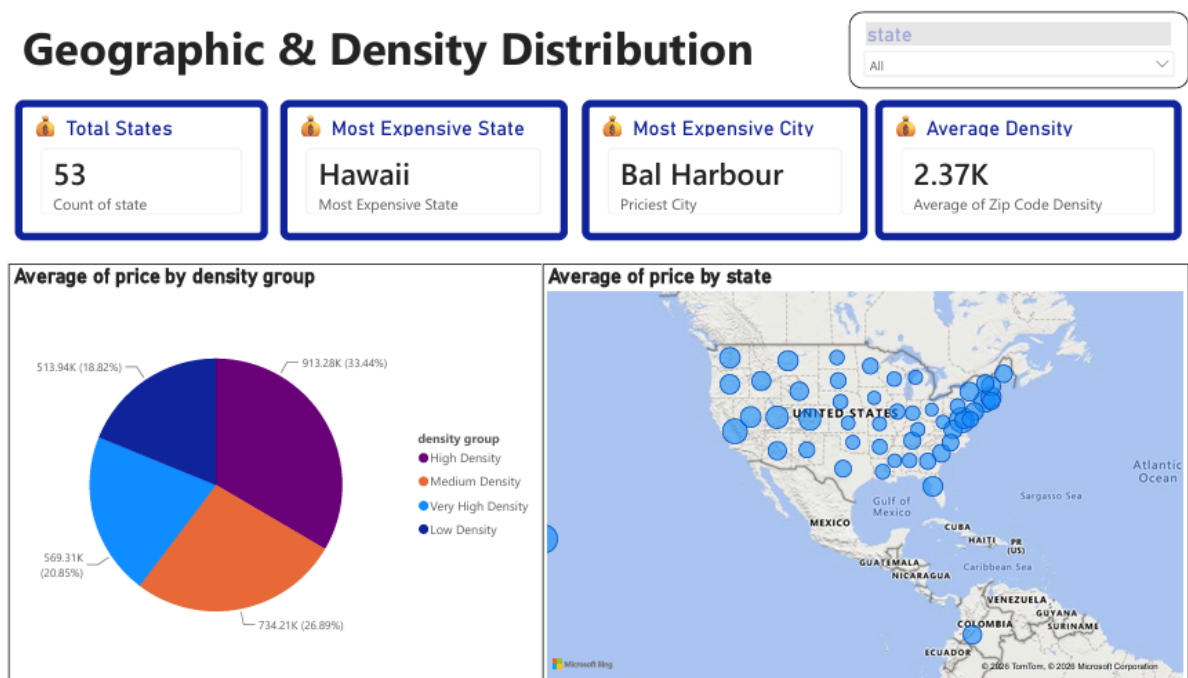


Figure 3: Geographic and Density Distribution Dashboard

Objectives

This dashboard aims to explore the geographic distribution of housing prices by state and city, and to study how density affects property values. It will help stakeholders determine location-based investment opportunities by plotting price differences in a geographic pattern and contrasting average prices by density categories.

Dashboard Description

Figure 3 presents the Geographic & Density Distribution dashboard, which provides a spatial and density-based perspective on housing prices. The dashboard displays four KPIs: Total States Covered (53), Most Expensive State (Hawaii), Most Expensive City (Bal Harbour), and Average Zip Code Density (2.37K).

The Average Price by Density Group pie chart breaks down average housing prices across four density classifications. This distribution reveals that residential density does not follow a linear relationship with price, suggesting that neighbourhood type and property composition within each density group play a significant role in determining value.

The Average Price by State map provides a geographic overview of price distribution across the United States, allowing users to identify high-value regions and regional pricing clusters visually. A state slicer enables users to filter both visuals interactively for more targeted geographic analysis.

4.0 Key Insights

Based on the dashboards, the key insights found are:

1. Housing affordability varies significantly across locations. Cities such as Curtis Bay, Hanford and Grosse Pointe have lower affordability ratios, indicating that housing prices are more aligned with local household income levels. These locations may represent attractive and affordable investment opportunities.
2. Several states exhibit low housing affordability. California, New York, and Florida appear among the least affordable states, where housing prices are substantially higher relative to household income. This indicates higher acquisition costs and reduced affordability in these markets.
3. Market demand is concentrated in medium and small house sizes. Medium and Small houses make up the majority of properties, while Luxury houses, though limited in number, command the highest average prices.
4. Property condition affects market value. Houses in better condition generally command higher average prices. It means that well-maintained properties are more valuable in the housing market.
5. The number of bedrooms and bathrooms shows a weaker and less consistent relationship with price compared to house size and condition. This indicates it is not a strong standalone predictor of property value.
6. Residential density does not directly correlate with higher prices. High Density areas record the highest average house price at 913.28K (33.44%), while Very High Density areas have the lowest at 513.94K (18.62%). This suggests that extremely dense urban zones may consist of smaller, more affordable multi-unit housing, whereas high density suburban areas tend to feature larger and more premium properties.
7. Geographic location significantly influences housing prices, with Hawaii identified as the most expensive state and Bal Harbour as the most expensive city. This indicates that coastal and island markets command a substantial price premium, reflecting the influence of geographic desirability, limited land supply, and tourism-driven demand on property values.

5.0 Business Recommendations

Based on the key insights, the following business recommendations are proposed:

1. Prioritise investment opportunities in cities with lower affordability ratios, as these locations offer more attractive entry points and better affordability relative to local income levels.
2. Carefully evaluate investments in states with high affordability ratios such as California and New York, as higher acquisition costs may increase investment risk and reduce affordability.
3. Focus on developing medium and small houses, as they represent the largest share of market demand, while including a limited number of luxury houses to capture higher profit margins.
4. Encourage property maintenance and renovations before selling, as improving a property's condition can increase its market value and attract more potential buyers.
5. Prioritise investment in High Density areas when targeting premium properties, as they record the highest average price at 913.28K compared to other density groups, offering stronger returns for premium market segments.
6. Approach geographically premium markets such as Hawaii and Bal Harbour with careful financial planning, as limited land availability and high coastal desirability drive prices significantly higher, requiring thorough capital assessment before entry.